Spreadsheets Homework

Status	Done
Assign	Pasit Promchawee
Classes	Live 04 - Spreadsheets

▼ Homework Sheet:

Copy of Bootcamp 12 Google Sheets

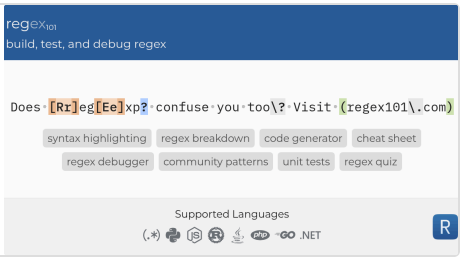
https://docs.google.com/spreadsheets/d/1juECyi0v23nrutJm3o6Har3zjOTjsK0Gu7EXNReZ2Iw/edit?usp=sharing

▼ Usefulness

regex101: build, test, and debug regex

Regular expression tester with syntax highlighting, explanation, cheat sheet for PHP/PCRE, Python, GO, JavaScript, Java, C#/.NET, Rust.

https://regex101.com



List of homework

- Create Conditional Formatting
- Project ID Card Parser
  - ID
  - Gender
  - Firstname
  - Lastname
  - DOB
  - Age

- Address
- Zipcode
- Expired Date
- Expired This Year

Create Dynamic Query (New)

Convert TH date to EN date (New)

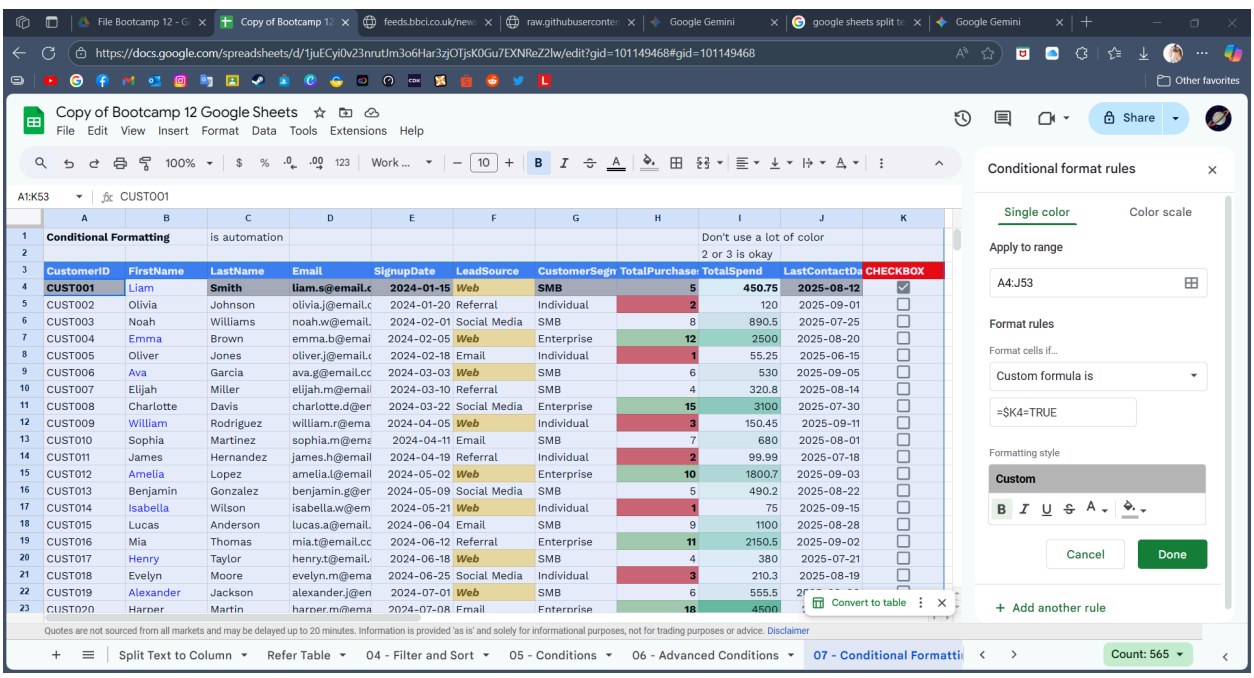
- Method 1 - Separate Function
- Method 2 - Combine All Functions

Share your note (New)

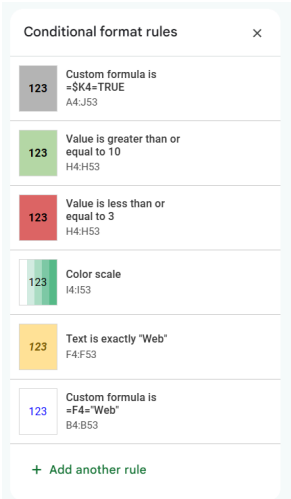
Create Conditional Formatting

- ☑ ~~Create new column as “Checkbox”~~
- ☑ ~~Highlight the entire row when checkbox is TRUE (Checked)~~

First, highlight the range that you want to apply then go to conditional formatting

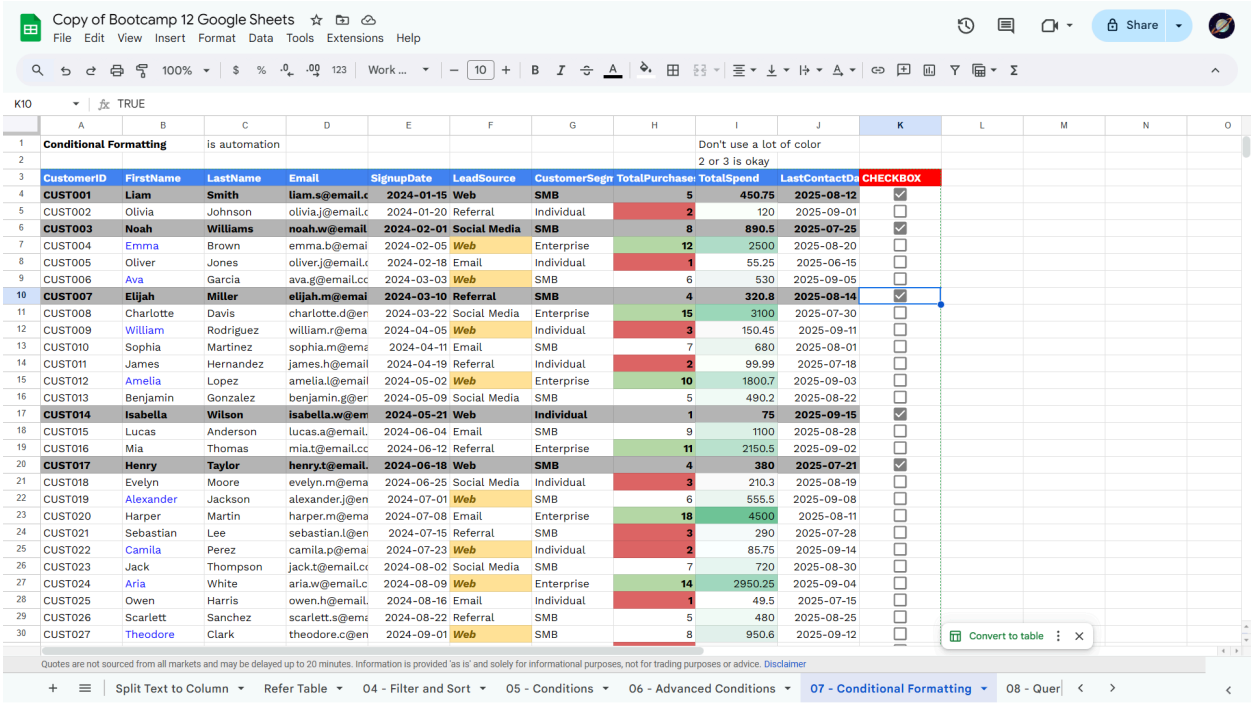


Use “Custom formula is” then `=$K4=TRUE` .



And then change the rule priority to the top. This will overlap any existing format.

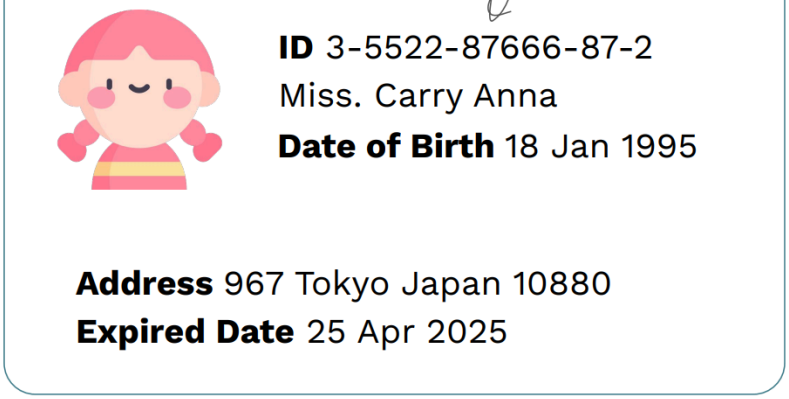
And here the result:



Project ID Card Parser

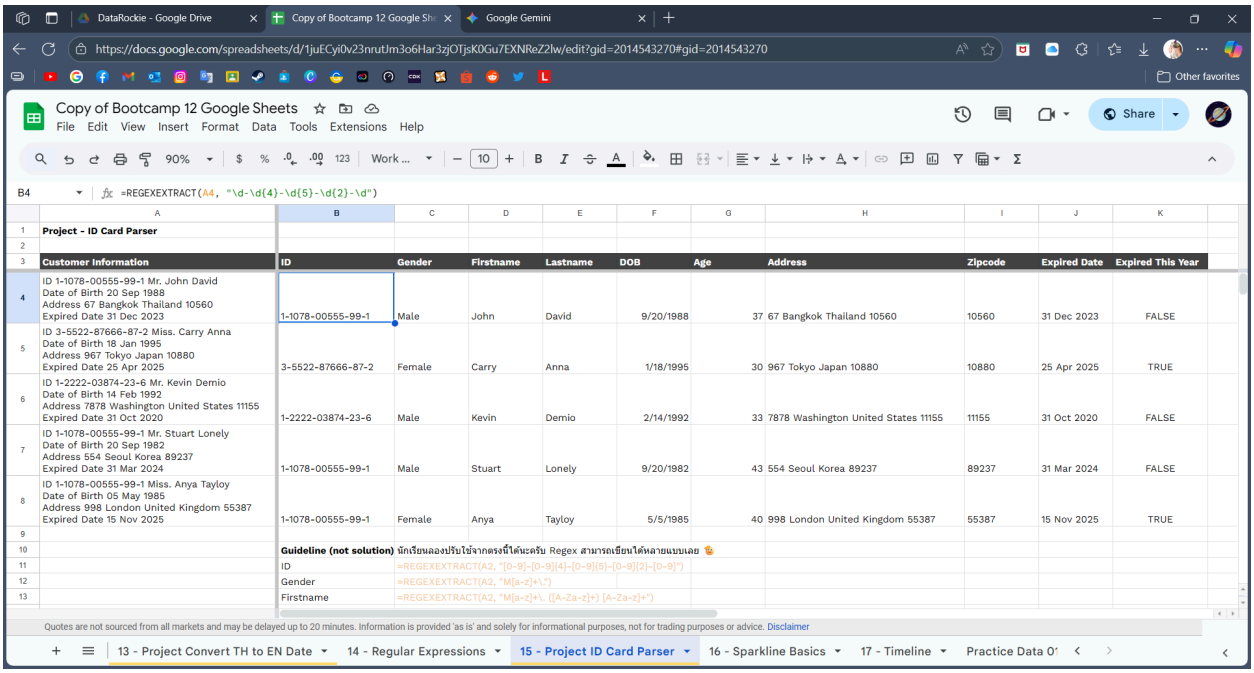
Project - ID Card Parser

We will extract information from this card



Extract the information with regular expression (REGEX).

- [x] ~~ID~~
- [x] ~~Gender~~
- [x] ~~Firstname~~
- [x] ~~Lastname~~
- [x] ~~DOB (Date of Birth)~~
- [x] ~~Age~~
- [x] ~~Address~~
- [x] ~~Zipcode~~
- [x] ~~Expired Date~~
- [x] ~~Expired This Year~~

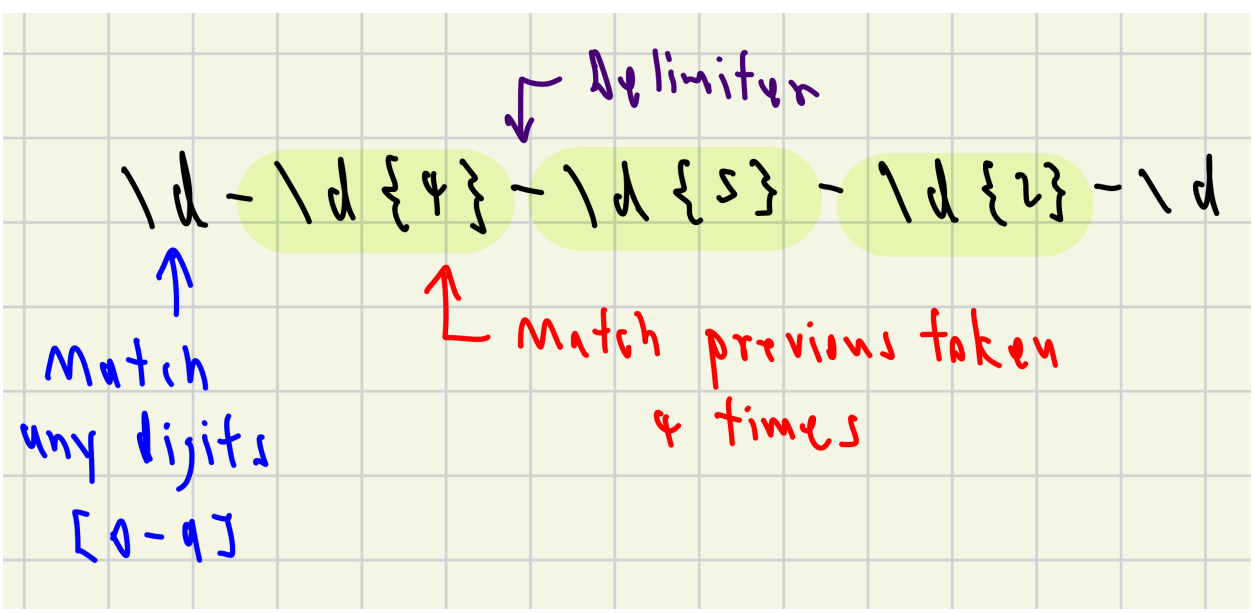


ID

```
=REGEXEXTRACT(A4, "[0-9]-[0-9]{4}-[0-9]{5}-[0-9]{2}-[0-9]")
=REGEXEXTRACT(A4, "\d-\d{4}-\d{5}-\d{2}-\d")
```

If `\d` is X

X-XXXX-XXXXX-XX-X

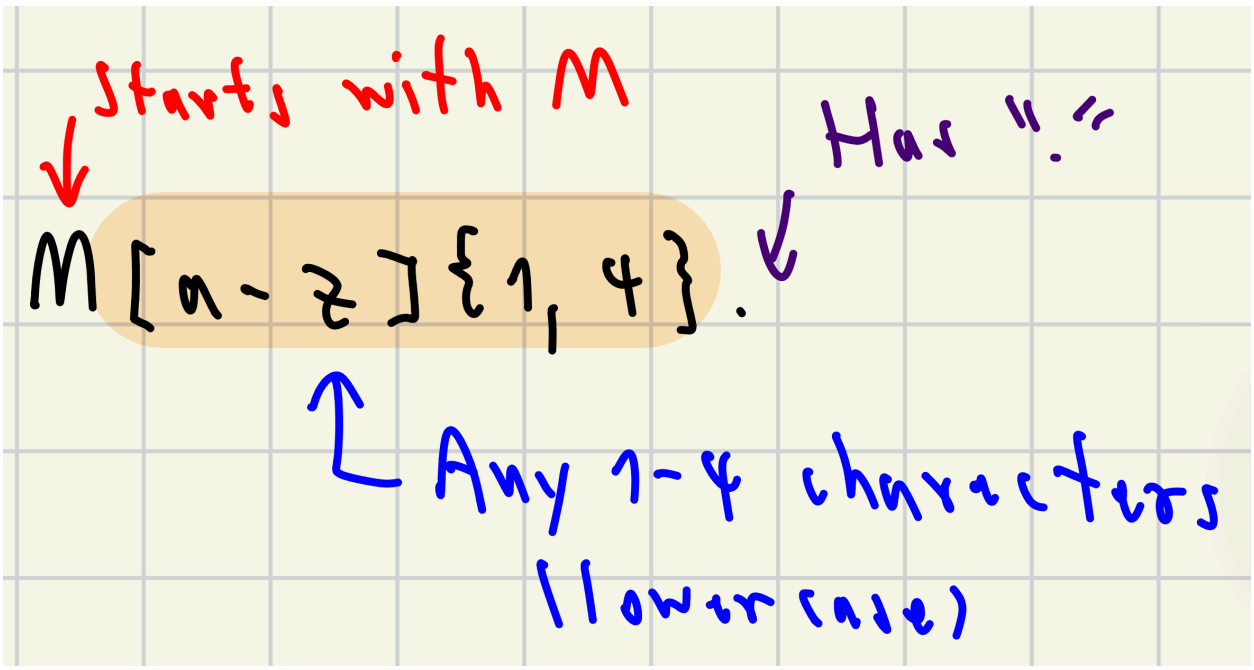


Gender

```
=LET(PREFIX, REGEXEXTRACT(A4, "M[a-z]{1,4}."),
IF(PREFIX = "Mr.", "Male", "Female"))
```

Text would be like M___.

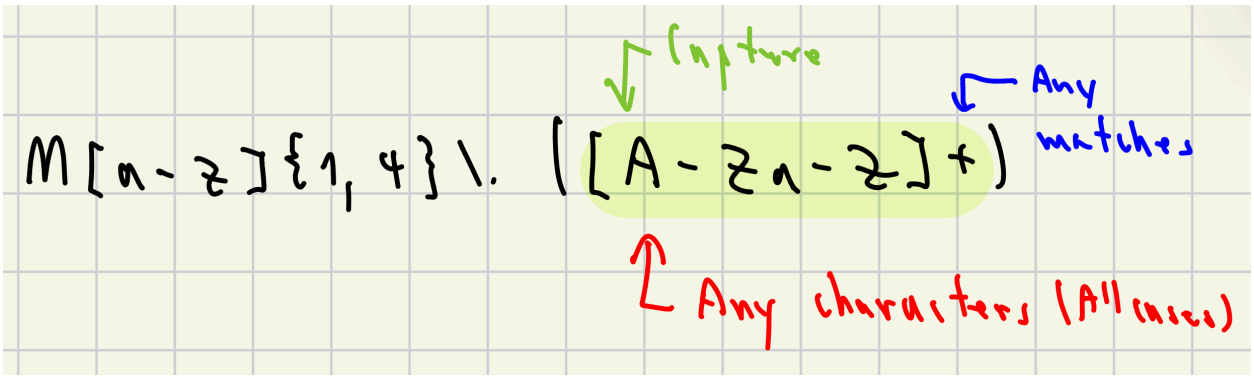
Within the _ can be 1-4 characters (lowercase) and end with period "."



Firstname

```
=REGEXEXTRACT(A4, "M[a-z]{1,4}\. ([A-Za-z]+)")
```

Focus in parenthesis, choose any characters afterward.



Lastname

```
=REGEXEXTRACT(A4, "M[a-z]{1,4}\. [A-Za-z]+ ([A-Za-z]+)")
```

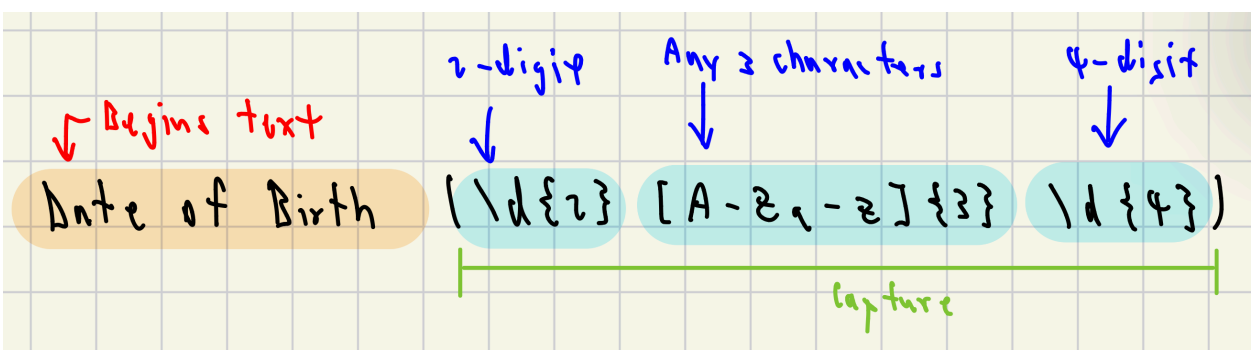
Focus in parenthesis, choose any characters afterward.

(The same as above, just adds 1 more block)

DOB

```
=REGEXEXTRACT(A4,"Date of Birth (\d{2} [A-Za-z]+ \d{4})")
=LET(
REGEXDAY, REGEXEXTRACT(A4, "Date of Birth (\d{2})"),
REGEXMONTH, REGEXEXTRACT(A4, "Date of Birth \d{2} ([A-Za-z]{3})"),
REGEXYEAR, REGEXEXTRACT(A4, "Date of Birth \d{2} [A-Za-z]{3} (\d{4})"),
DATE(VALUE(REGEXYEAR), MONTH(REGEXMONTH&" 1"), VALUE(REGEXDAY)))
```

Extract each day, month and year. And then convert to date format.



Age

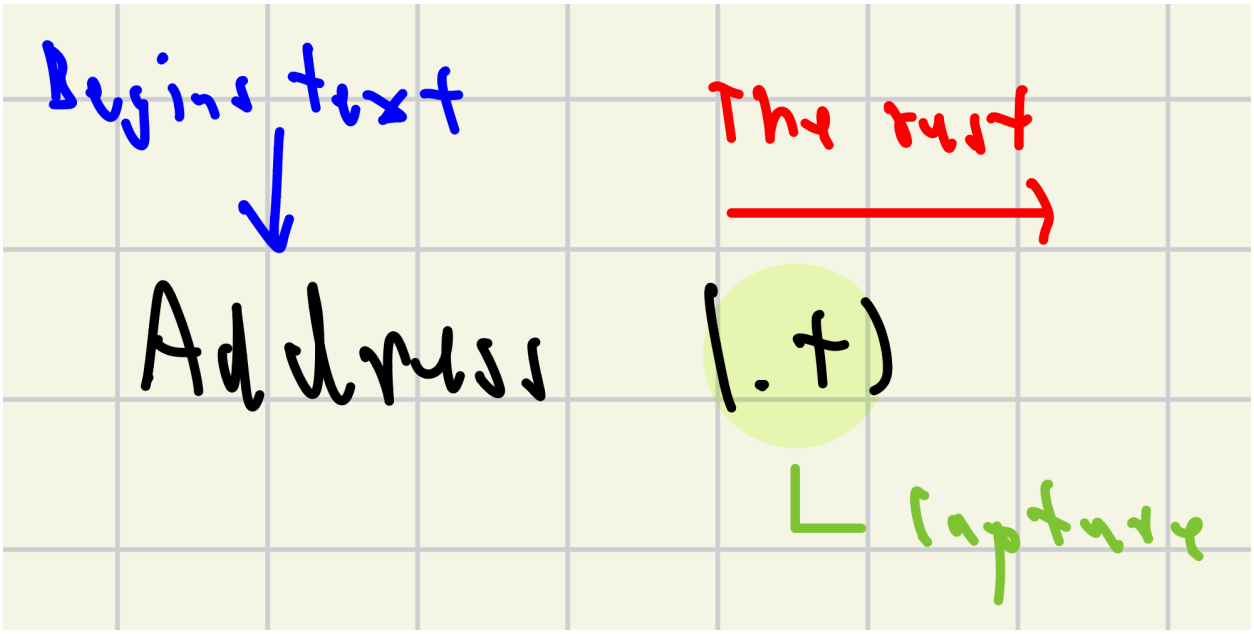
```
=DATEDIF(F4,TODAY(),"Y")
```

Find the different year between today and DOB.

Address

```
=REGEXEXTRACT(A4, "Address (.+)")
```

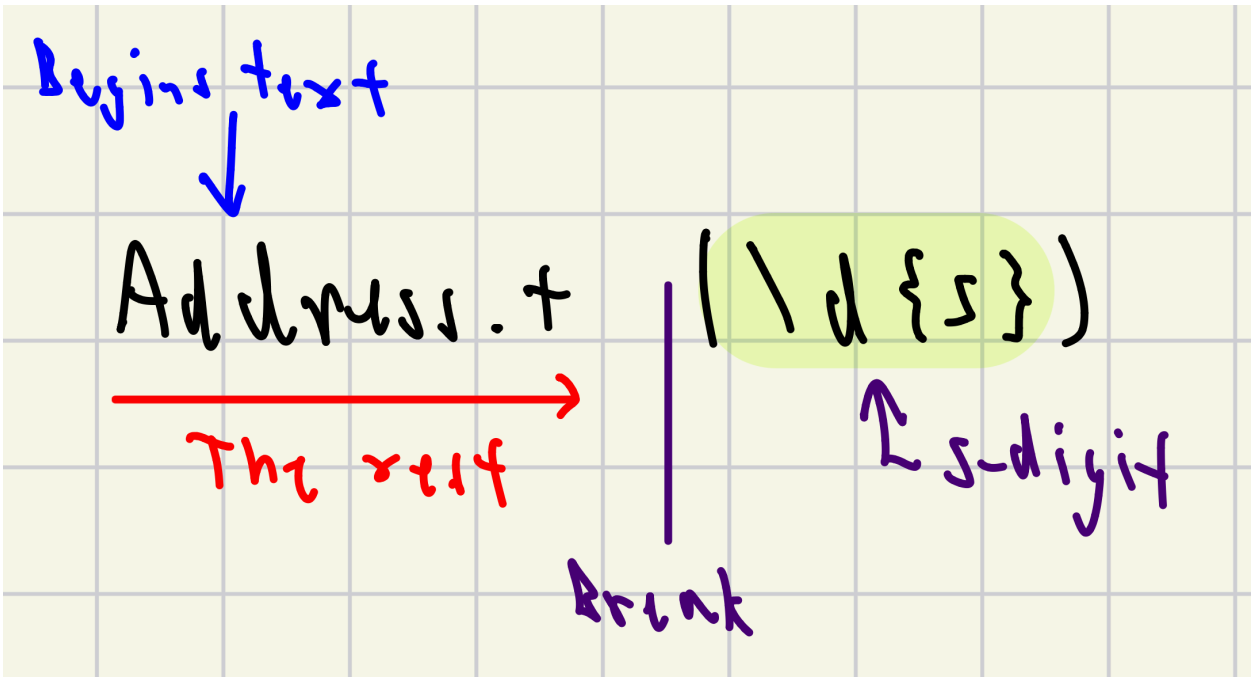
Extract the entire address.



Zipcode

```
=REGEXEXTRACT(A4, "Address.+ (\d{5})")
```

Extract only 5-digit zip code.



Expired Date

```
=REGEXEXTRACT(A4, "Expired Date (.+)")
```

Extract the expired date.

(The same method as Address.)

Expired This Year

```
=IF(RIGHT(J4, 4) = "2025", TRUE, FALSE)
```

If year is equal 2025 return TRUE.

Create Dynamic Query (New)

I'm using 3 drop downs:

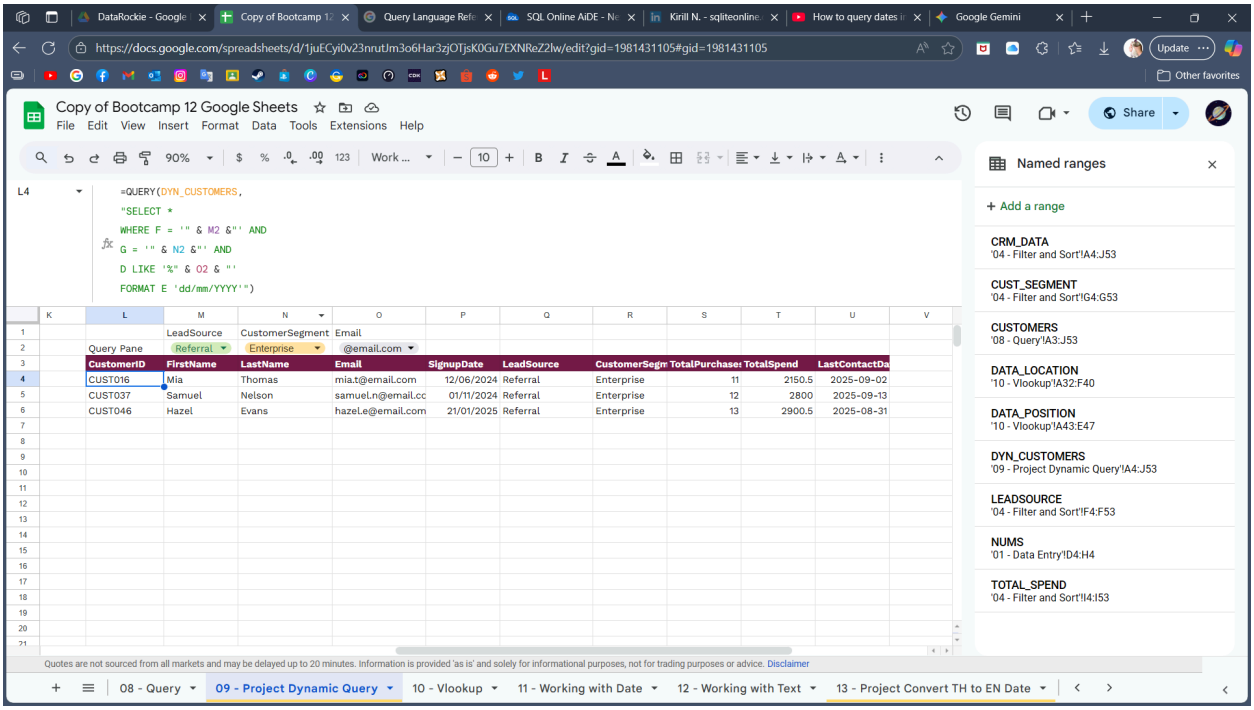
- LeadSource
- CustomerSegment

- Email (Literally, it only has `@email.com`)

And this is how I write `QUERY()`

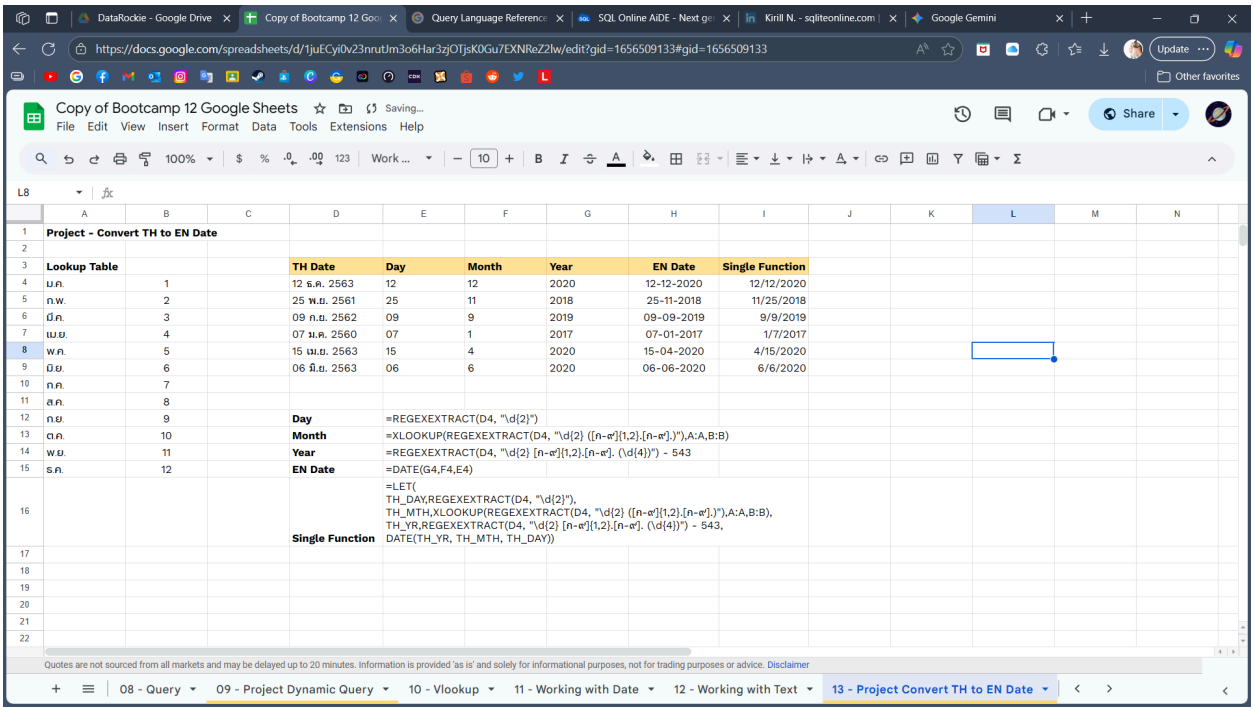
```
=QUERY(DYN_CUSTOMERS,
"SELECT *
WHERE F = '" & M2 &"' AND
G = '" & N2 &"' AND
D LIKE '%" & O2 & "'
FORMAT E 'dd/mm/YYYY'")
```

```
SELECT A, B, C, D, strftime('%d/%m/%Y', E) E, F, I, J
FROM DYN_CUSTOMERS
WHERE F = 'Referral' AND
G = 'Enterprise' AND
D LIKE '&@email.com'
```



Convert TH date to EN date (New)

The task is transform Thai date to English date.



Method 1 - Separate Function

1. Get `DAY`

```
=REGEXEXTRACT(D4, "\d{2}") // Extract first 2 digits
```

2. Get `Month`

```
=XLOOKUP(REGEXEXTRACT(D4, "\d{2} ([ก-ฮ]{1,2}.[ก-ฮ].)"),A:A,B:B)
// Extract the short TH month name and then lookup to return month number
```

3. Get `Year`

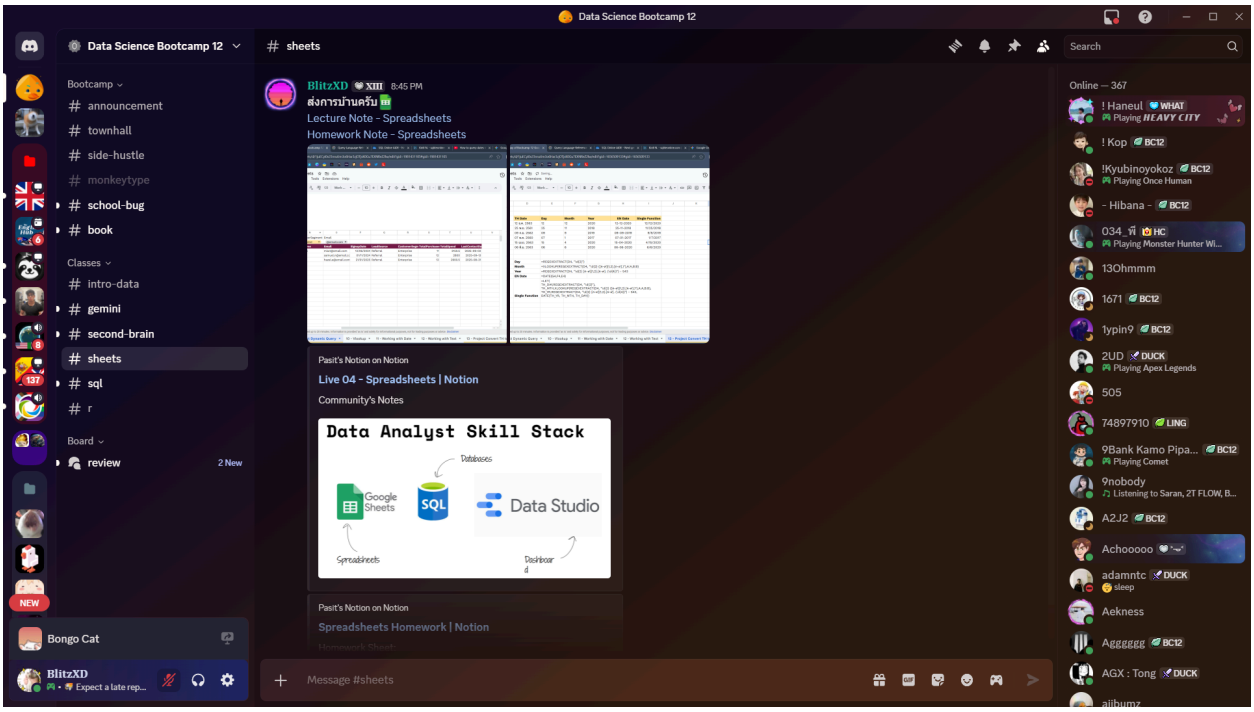
```
=REGEXEXTRACT(D4, "\d{2} [ก-ฮ]{1,2}.[ก-ฮ]. (\d{4})") - 543
// Extract last 4 digits and minus 543 to convert as AD year.
```

4. Combine with `DATE()`

```
=DATE(G4,F4,E4)
```

Method 2 - Combine All Functions

I use `LET()` to define variable like doing in programming language.

```
=LET(
TH_DAY,REGEXEXTRACT(D4, "\d{2}"), // Get day
TH_MTH,XLOOKUP(REGEXEXTRACT(D4, "\d{2} ([ก-ฮ]{1,2}.[ก-ฮ].)"),A:A,B:B), // Get month
TH_YR,REGEXEXTRACT(D4, "\d{2} [ก-ฮ]{1,2}.[ก-ฮ]. (\d{4})") - 543, // Get year
DATE(TH_YR, TH_MTH, TH_DAY)) // Combine with DATE()
```

Share your note (New)